

Sublinear Regret of LSVI-UCB on Linear MDPs

1 Preliminaries

Setup and notation. We study an episodic, finite-horizon Markov decision process $\text{MDP}(\mathcal{S}, \mathcal{A}, H, P, r)$ with state space $\mathcal{S}$, action space $\mathcal{A}$, horizon H , transition kernels $P = \{P_h\}_{h \in [H]}$, and reward functions $r = \{r_h\}_{h \in [H]}$ with $r_h(x, a) \in [0, 1]$. An agent interacts for K episodes; we write $T := KH$ for the total number of steps and $[n] := \{1, \dots, n\}$. For a policy $\pi = \{\pi_h\}_{h \in [H]}$, the value and action-value functions are

$$V_h^\pi(x) = \mathbb{E}\left[\sum_{h'=h}^H r_{h'}(x_{h'}, a_{h'}) \mid x_h = x\right], \quad Q_h^\pi(x, a) = r_h(x, a) + (P_h V_{h+1}^\pi)(x, a),$$

where $(P_h g)(x, a) := \mathbb{E}_{x' \sim P_h(\cdot | x, a)}[g(x')]$ and $V_{H+1}^\pi \equiv 0$. We write V_h^*, Q_h^* for the optimal value functions and $\mathbb{T}_h g(x, a) := r_h(x, a) + (P_h g)(x, a)$ for the one-step Bellman backup of a function $g : \mathcal{S} \rightarrow \mathbb{R}$. Throughout, $c, C, C_1, C_2, \dots$ denote universal positive constants whose values may change from line to line and depend only on quantities specified in each statement's hypotheses; any constant with a problem-dependent meaning is written with an explicit subscript (e.g. C_β).

Regret. At episode k the agent plays policy π^k starting from a state x_1^k . The cumulative regret over K episodes is

$$\text{Regret}(K) := \sum_{k=1}^K [V_1^*(x_1^k) - V_1^{\pi^k}(x_1^k)]. \quad (1)$$

Assumption 1.1 (Linear MDP). There exists a known feature map $\phi : \mathcal{S} \times \mathcal{A} \rightarrow \mathbb{R}^d$ such that for every $h \in [H]$ there exist d unknown signed measures $\mu_h = (\mu_h^{(1)}, \dots, \mu_h^{(d)})$ over $\mathcal{S}$ and an unknown vector $\theta_h \in \mathbb{R}^d$ with

$$P_h(\cdot | x, a) = \langle \phi(x, a), \mu_h(\cdot) \rangle, \quad r_h(x, a) = \langle \phi(x, a), \theta_h \rangle, \quad \forall (x, a) \in \mathcal{S} \times \mathcal{A}.$$

We normalize $\|\phi(x, a)\| \leq 1$ for all (x, a) and $\max\{\|\mu_h(\mathcal{S})\|, \|\theta_h\|\} \leq \sqrt{d}$ for all $h \in [H]$, where $\mu_h(\mathcal{S}) := \int_{\mathcal{S}} d\mu_h(x') \in \mathbb{R}^d$.

Remark 1.2. Assumption 1.1 is the standard linear-MDP model of [2]. It strictly generalizes the tabular MDP (take $\phi(x, a) = e_{(x,a)}$, $d = |\mathcal{S}||\mathcal{A}|$) and the simplex feature model, yet the transition kernel retains infinitely many degrees of freedom because μ_h is unknown. The normalization $\|\phi\| \leq 1$, $\|\theta_h\|, \|\mu_h(\mathcal{S})\| \leq \sqrt{d}$ is a scaling convention rather than a restriction: any bounded feature map can be rescaled to meet it.

The LSVI-UCB algorithm. We analyze Least-Squares Value Iteration with an Upper-Confidence-Bound bonus ([2]). Fix a regularizer $\lambda > 0$ and a bonus radius $\beta > 0$. At episode k , for $h = H, H-1, \dots, 1$ the algorithm forms the Gram matrix and ridge weights

$$\Lambda_h^k = \sum_{\tau=1}^{k-1} \phi(x_h^\tau, a_h^\tau) \phi(x_h^\tau, a_h^\tau)^\top + \lambda I, \quad (2)$$

$$w_h^k = (\Lambda_h^k)^{-1} \sum_{\tau=1}^{k-1} \phi(x_h^\tau, a_h^\tau) [r_h(x_h^\tau, a_h^\tau) + V_{h+1}^k(x_{h+1}^\tau)], \quad (3)$$

and sets the optimistic action-value and value functions

$$Q_h^k(x, a) = \min \left\{ \left\langle w_h^k, \phi(x, a) \right\rangle + \beta \|\phi(x, a)\|_{(\Lambda_h^k)^{-1}}, H \right\}, \quad (4)$$

$$V_h^k(x) = \max_{a \in \mathcal{A}} Q_h^k(x, a), \quad (5)$$

with $V_{H+1}^k \equiv 0$, where $\|u\|_M := (u^\top M u)^{1/2}$ is the M -weighted (Mahalanobis) norm. The agent then plays the greedy policy π^k , $a_h^k = \arg \max_a Q_h^k(x_h^k, a)$. Let $\iota := \log(2dT/\delta)$ for the target failure probability $\delta \in (0, 1)$; the analysis uses $\lambda = 1$ and $\beta = C_\beta dH \sqrt{\iota}$ for a universal constant C_β fixed in Lemma 2.2.

Remark 1.3. We work on the filtration $\{\mathcal{F}_{k,h}\}$ generated by all transitions observed up to and including step h of episode k . The pair (Λ_h^k, w_h^k) is $\mathcal{F}_{k-1,H}$ -measurable: it depends only on the first $k-1$ episodes. Consequently V_h^k is determined before the fresh transition at episode k is revealed, a fact used in the martingale argument of Section 5.

The following structural fact is the foundation of the analysis: in a linear MDP the Bellman backup of *any* bounded function is linear in the features.

Fact 1.4 (Linearity of the Bellman backup). Under Assumption 1.1, for every $h \in [H]$ and every measurable $g : \mathcal{S} \rightarrow [0, H]$ there is a vector $w \in \mathbb{R}^d$ with

$$\mathbb{T}_h g(x, a) = \langle \phi(x, a), w \rangle, \quad w = \theta_h + \int_{\mathcal{S}} g(x') d\mu_h(x'), \quad \|w\| \leq 2H\sqrt{d}. \quad (6)$$

In particular, for any policy π there exist weights w_h^π with $Q_h^\pi(x, a) = \langle \phi(x, a), w_h^\pi \rangle$ and $\|w_h^\pi\| \leq 2H\sqrt{d}$.

Proof. We compute the backup directly and then bound its weight vector. For the identity,

$$\begin{aligned} \mathbb{T}_h g(x, a) &= r_h(x, a) + \int_{\mathcal{S}} g(x') P_h(dx' | x, a) \\ &= \langle \phi(x, a), \theta_h \rangle + \int_{\mathcal{S}} g(x') \langle \phi(x, a), d\mu_h(x') \rangle \\ &= \langle \phi(x, a), \theta_h \rangle + \left\langle \phi(x, a), \int_{\mathcal{S}} g(x') d\mu_h(x') \right\rangle \\ &= \left\langle \phi(x, a), \theta_h + \int_{\mathcal{S}} g(x') d\mu_h(x') \right\rangle, \end{aligned}$$

where the first step is the definition of $\mathbb{T}_h$, the second substitutes Assumption 1.1 for both r_h and P_h , the third pulls the x' -independent vector $\phi(x, a)$ out of the integral by linearity of the inner product, and the last collects terms. This proves Eq. (6) with $w = \theta_h + \int g d\mu_h$. For the norm,

$$\|w\| \leq \|\theta_h\| + \left\| \int_{\mathcal{S}} g(x') d\mu_h(x') \right\| \leq \sqrt{d} + H \|\mu_h(\mathcal{S})\| \leq \sqrt{d} + H\sqrt{d} \leq 2H\sqrt{d},$$

where the first step is the triangle inequality, the second uses $0 \leq g \leq H$ together with $\int d\mu_h = \mu_h(\mathcal{S})$, the third applies the normalization $\|\theta_h\|, \|\mu_h(\mathcal{S})\| \leq \sqrt{d}$ from Assumption 1.1, and the last uses $H \geq 1$. Applying the identity with $g = V_{h+1}^\pi \in [0, H]$ and recalling $Q_h^\pi = \mathbb{T}_h V_{h+1}^\pi$ yields the final claim. This completes the proof. $\square$

2 Concentration and the successful event

We first bound the magnitude of the ridge weights produced by LSVI-UCB, then use a self-normalized concentration argument to define a high-probability event on which least-squares value iteration tracks the Bellman backup.

Lemma 2.1 (Bounded weights). *Under Assumption 1.1 with $\lambda = 1$, for every $(k, h) \in [K] \times [H]$ the LSVI-UCB weight w_h^k of Eq. (3) satisfies $\|w_h^k\| \leq 2H\sqrt{dk}$.*

Proof. We bound $|v^\top w_h^k|$ for an arbitrary unit vector v and take the supremum. Writing $\phi_h^\tau := \phi(x_h^\tau, a_h^\tau)$ and $y_h^\tau := r_h(x_h^\tau, a_h^\tau) + V_{h+1}^k(x_{h+1}^\tau) \in [0, H]$,

$$\begin{aligned} |v^\top w_h^k| &= \left| \sum_{\tau=1}^{k-1} v^\top (\Lambda_h^k)^{-1} \phi_h^\tau y_h^\tau \right| \\ &\leq H \sum_{\tau=1}^{k-1} |v^\top (\Lambda_h^k)^{-1} \phi_h^\tau| \\ &\leq H \left(\sum_{\tau=1}^{k-1} v^\top (\Lambda_h^k)^{-1} v \right)^{1/2} \left(\sum_{\tau=1}^{k-1} (\phi_h^\tau)^\top (\Lambda_h^k)^{-1} \phi_h^\tau \right)^{1/2} \\ &\leq H \cdot \|v\| \cdot (dk)^{1/2} = H\sqrt{dk}, \end{aligned}$$

where the first step is Eq. (3), the second uses $|y_h^\tau| \leq H$ and the triangle inequality, the third is the Cauchy–Schwarz inequality in the inner product $\langle u, u' \rangle_{(\Lambda_h^k)^{-1}} = u^\top (\Lambda_h^k)^{-1} u'$, and the fourth uses $v^\top (\Lambda_h^k)^{-1} v \leq \|v\|^2 / \lambda = \|v\|^2$ (since $\lambda_{\min}(\Lambda_h^k) \geq \lambda = 1$) together with $\sum_{\tau=1}^{k-1} (\phi_h^\tau)^\top (\Lambda_h^k)^{-1} \phi_h^\tau \leq d$ (Lemma 4.1) and $\|v\| = 1$ (so the first factor is $(\sum_{\tau=1}^{k-1} \|v\|^2)^{1/2} = \sqrt{k-1} \leq \sqrt{k}$). Taking the supremum over $\|v\| = 1$ and using $\|w_h^k\| = \sup_{\|v\|=1} |v^\top w_h^k|$ gives $\|w_h^k\| \leq H\sqrt{dk} \leq 2H\sqrt{dk}$. This completes the proof. $\square$

The core statistical guarantee is the following self-normalized bound on the fluctuation of the empirical Bellman backup. We define the successful event $\mathcal{E}$ here; all later derivations are carried out on $\mathcal{E}$.

Lemma 2.2 (Self-normalized concentration; successful event). *Suppose Assumption 1.1 holds and $\lambda = 1$. There exist universal constants $C_\beta > 0$ and $C > 0$ such that, setting $\beta = C_\beta dH\sqrt{\iota}$ with $\iota = \log(2dT/\delta)$, the event*

$$\mathcal{E} := \left\{ \forall (k, h) \in [K] \times [H] : \left\| \sum_{\tau=1}^{k-1} \phi_h^\tau [V_{h+1}^k(x_{h+1}^\tau) - (P_h V_{h+1}^k)(x_h^\tau, a_h^\tau)] \right\|_{(\Lambda_h^k)^{-1}} \leq C dH\sqrt{\iota} \right\}. \quad (7)$$

satisfies $\mathbb{P}[\mathcal{E}] \geq 1 - \delta/2$.

Proof. The argument is the self-normalized tail bound for vector-valued martingales of [1], combined with a covering-number union bound over the class of value functions V_{h+1}^k can equal; we record its structure and the constant accounting.

Fix (k, h) and abbreviate $\phi_h^\tau := \phi(x_h^\tau, a_h^\tau)$. For a *fixed* function $V : \mathcal{S} \rightarrow [0, H]$, the increments $\varepsilon_h^\tau(V) := V(x_{h+1}^\tau) - (P_h V)(x_h^\tau, a_h^\tau)$ form a martingale-difference sequence with respect to $\{\mathcal{F}_{\tau, h}\}$ with $\mathbb{E}[\varepsilon_h^\tau(V) \mid \mathcal{F}_{\tau, h-1}] = 0$ and $|\varepsilon_h^\tau(V)| \leq H$, hence conditionally H -sub-Gaussian; the regressors ϕ_h^τ are $\mathcal{F}_{\tau, h-1}$ -measurable, hence $\{\mathcal{F}_{\tau, h-1}\}$ -predictable, so the hypotheses of Lemma 4.4 are met. By the self-normalized bound of [1] (recorded in Lemma 4.4), for any fixed V and any $\delta' \in (0, 1)$, with probability at least $1 - \delta'$,

$$\begin{aligned} \left\| \sum_{\tau=1}^{k-1} \phi_h^\tau \varepsilon_h^\tau(V) \right\|_{(\Lambda_h^k)^{-1}}^2 &\leq 2H^2 \log \left(\frac{\det(\Lambda_h^k)^{1/2} \det(\lambda I)^{-1/2}}{\delta'} \right) \\ &\leq 2H^2 \left(\frac{d}{2} \log \frac{\lambda+k}{\lambda} + \log \frac{1}{\delta'} \right), \end{aligned} \quad (8)$$

where the first step is Lemma 4.4 with $\sigma = H$ and the second bounds $\det(\Lambda_h^k) \leq (\lambda + k)^d$ (the eigenvalues of Λ_h^k sum to at most $\lambda d + k$, so their product is at most $(\lambda + k/d)^d \leq (\lambda + k)^d$ by AM–GM) and $\det(\lambda I) = \lambda^d$.

The function V_{h+1}^k in Eq. (7) is *not* fixed: it depends on the data through $(w_{h+1}^k, \Lambda_{h+1}^k)$. By Eq. (4) and Eq. (5) together with Lemma 2.1, every such V_{h+1}^k lies in the parametric class

$$\mathcal{V} := \left\{ x \mapsto \min \left\{ \max_a \left[\langle w, \phi(x, a) \rangle + \beta \|\phi(x, a)\|_{A^{-1}} \right], H \right\} : \|w\| \leq 2H\sqrt{dK}, A \succeq I \right\}.$$

Let $\mathcal{V}_\epsilon$ be a minimal ϵ -cover of $\mathcal{V}$ in the supremum norm. A volumetric estimate over the parameters (w, A) ([2], Lemma D.6) gives

$$\log |\mathcal{V}_\epsilon| \leq C_1 d^2 \log \left(1 + \frac{HdK}{\epsilon\lambda} \right). \quad (9)$$

Apply Eq. (8) with $\delta' = \delta/(2HK|\mathcal{V}_\epsilon|)$ and union-bound over $\mathcal{V}_\epsilon$ and over all (k, h) , so that the fixed- V bound holds simultaneously for every $\bar{V} \in \mathcal{V}_\epsilon$. The realized $V_{h+1}^k \in \mathcal{V}$ need not lie in $\mathcal{V}_\epsilon$, so we transfer to its nearest cover element $\bar{V}$ with $\|V_{h+1}^k - \bar{V}\|_\infty \leq \epsilon$: since ε_h^τ is 1-Lipschitz in V under $\|\cdot\|_\infty$ and $\|\sum_\tau \phi_h^\tau \cdot\|_{(\Lambda_h^k)^{-1}}$ is a seminorm, the discretization correction is additive,

$$\left\| \sum_{\tau=1}^{k-1} \phi_h^\tau \varepsilon_h^\tau(V_{h+1}^k) \right\|_{(\Lambda_h^k)^{-1}} \leq \left\| \sum_{\tau=1}^{k-1} \phi_h^\tau \varepsilon_h^\tau(\bar{V}) \right\|_{(\Lambda_h^k)^{-1}} + 2\epsilon\sqrt{k},$$

the second term bounding $\|\sum_\tau \phi_h^\tau [\varepsilon_h^\tau(V_{h+1}^k) - \varepsilon_h^\tau(\bar{V})]\|_{(\Lambda_h^k)^{-1}} \leq 2\epsilon(\sum_\tau (\phi_h^\tau)^\top (\Lambda_h^k)^{-1} \phi_h^\tau)^{1/2} \leq 2\epsilon\sqrt{d} \leq 2\epsilon\sqrt{k}$ via Cauchy–Schwarz and Lemma 4.1. With $\epsilon = dH/K$ this correction is at most $2dH\sqrt{K}/K \leq 2dH$, lower order than the radius below. Hence, with probability at least $1 - \delta/2$, simultaneously for all (k, h) ,

$$\begin{aligned} \left\| \sum_{\tau=1}^{k-1} \phi_h^\tau \varepsilon_h^\tau(V_{h+1}^k) \right\|_{(\Lambda_h^k)^{-1}}^2 &\leq 2 \cdot 2H^2 \left(\frac{d}{2} \log(\lambda + k) + \log \frac{1}{\delta'} \right) + 2(2dH)^2 \\ &\leq 2H^2 \left(\frac{d}{2} \iota + C_2 d^2 \iota \right) \\ &\leq C^2 d^2 H^2 \iota, \end{aligned}$$

where the first inequality applies Eq. (8) to $\bar{V}$ together with the additive correction above (using $(a+b)^2 \leq 2a^2 + 2b^2$), the second substitutes $\log(1/\delta') \leq C_2 d^2 \iota$ from Eq. (9) (with $\iota = \log(2dT/\delta)$) absorbing the polynomial arguments $\log(1 + HdK/(\epsilon\lambda)) \lesssim \iota$, and the last collects the dominant d^2 term into a single universal C^2 . Taking square roots yields Eq. (7) with radius $C dH\sqrt{\iota}$, where C denotes this fixed event-radius constant (the same C that reappears in Lemma 3.1); choosing $C_\beta \geq C + 4$ in $\beta = C_\beta dH\sqrt{\iota}$ (the stronger of the two floors $C_\beta \geq C$ here and $C_\beta \geq C + 4$ in Lemma 3.1) makes β dominate this fluctuation. The constant C_β is thus a universal constant— independent of d, H, K, T, δ —whose existence and $dH\sqrt{\iota}$ scaling are exactly the exploration-bonus normalization of [2] (their Theorem 3.1, $\beta = c dH\sqrt{\iota}$): the constant is fixed only implicitly, by the self-consistency requirement that β exceed the event radius $C dH\sqrt{\iota}$ (equivalently their union-bound balance $c'\sqrt{\log 2 + \log(c_\beta + 1)} \leq c_\beta\sqrt{\log 2}$), and admits no closed-form numeric value; only the rate factors $dH\sqrt{\iota}$ are pinned. Hence $\mathbb{P}[\mathcal{E}] \geq 1 - \delta/2$. This completes the proof. $\square$

3 Bellman recursion and optimism

On the successful event $\mathcal{E}$ of Lemma 2.2 we now show two things: the LSVI estimate of the Bellman backup is accurate up to the bonus (Lemma 3.1), and consequently the optimistic value functions upper-bound the optimal ones (Lemma 3.2). Both are deterministic statements on $\mathcal{E}$.

Lemma 3.1 (One-step Bellman recursion). *Suppose Assumption 1.1 holds, $\lambda = 1$, and $\beta = C_\beta dH\sqrt{\iota}$ as in Lemma 2.2. On the event $\mathcal{E}$, for every policy π and every $(x, a, h, k) \in \mathcal{S} \times \mathcal{A} \times [H] \times [K]$,*

$$\left\langle \phi(x, a), w_h^k \right\rangle - Q_h^\pi(x, a) = P_h(V_{h+1}^k - V_{h+1}^\pi)(x, a) + \Delta_h^k(x, a), \quad |\Delta_h^k(x, a)| \leq \beta \|\phi(x, a)\|_{(\Lambda_h^k)^{-1}}. \quad (10)$$

Proof. By Fact 1.4, $Q_h^\pi = \langle \phi, w_h^\pi \rangle$ with $w_h^\pi = \theta_h + \int V_{h+1}^\pi d\mu_h$, so $\mathbb{T}_h V_{h+1}^\pi = \langle \phi, w_h^\pi \rangle$ and in particular $\langle \phi_h^\tau, w_h^\pi \rangle = r_h(x_h^\tau, a_h^\tau) + (P_h V_{h+1}^\pi)(x_h^\tau, a_h^\tau)$. Substituting the definition Eq. (3) of w_h^k and adding and subtracting λw_h^π inside the regularized normal equations,

$$\begin{aligned} w_h^k - w_h^\pi &= (\Lambda_h^k)^{-1} \sum_{\tau=1}^{k-1} \phi_h^\tau [r_h(x_h^\tau, a_h^\tau) + V_{h+1}^k(x_{h+1}^\tau)] - w_h^\pi \\ &= (\Lambda_h^k)^{-1} \left\{ -\lambda w_h^\pi + \sum_{\tau=1}^{k-1} \phi_h^\tau [V_{h+1}^k(x_{h+1}^\tau) - (P_h V_{h+1}^\pi)(x_h^\tau, a_h^\tau)] \right\} \\ &= \underbrace{-\lambda (\Lambda_h^k)^{-1} w_h^\pi}_{q_1} + \underbrace{(\Lambda_h^k)^{-1} \sum_{\tau=1}^{k-1} \phi_h^\tau [V_{h+1}^k(x_{h+1}^\tau) - (P_h V_{h+1}^k)(x_h^\tau, a_h^\tau)]}_{q_2} \\ &\quad + \underbrace{(\Lambda_h^k)^{-1} \sum_{\tau=1}^{k-1} \phi_h^\tau [(P_h V_{h+1}^k - P_h V_{h+1}^\pi)(x_h^\tau, a_h^\tau)]}_{q_3}, \end{aligned} \quad (11)$$

where the first step is Eq. (3), the second uses $\Lambda_h^k = \sum_{\tau} \phi_h^\tau (\phi_h^\tau)^\top + \lambda I$ to rewrite $w_h^\pi = (\Lambda_h^k)^{-1} \Lambda_h^k w_h^\pi$ and the normal-equation identity $\langle \phi_h^\tau, w_h^\pi \rangle = r_h^\tau + (P_h V_{h+1}^\pi)(x_h^\tau, a_h^\tau)$, and the third adds and subtracts $(P_h V_{h+1}^k)(x_h^\tau, a_h^\tau)$ inside the sum. Taking the inner product of Eq. (11) with $\phi(x, a)$ gives

$$\left\langle \phi(x, a), w_h^k - w_h^\pi \right\rangle = \langle \phi(x, a), q_1 \rangle + \langle \phi(x, a), q_2 \rangle + \langle \phi(x, a), q_3 \rangle. \quad (12)$$

We identify the three terms. For q_3 , since $P_h g(x, a) = \langle \phi(x, a), \int g d\mu_h \rangle$ by Assumption 1.1,

$$\begin{aligned} \langle \phi(x, a), q_3 \rangle &= \left\langle \phi(x, a), (\Lambda_h^k)^{-1} \sum_{\tau=1}^{k-1} \phi_h^\tau (\phi_h^\tau)^\top \int (V_{h+1}^k - V_{h+1}^\pi) d\mu_h \right\rangle \\ &= P_h(V_{h+1}^k - V_{h+1}^\pi)(x, a) - \lambda \left\langle \phi(x, a), (\Lambda_h^k)^{-1} \int (V_{h+1}^k - V_{h+1}^\pi) d\mu_h \right\rangle, \end{aligned}$$

where the first step writes $(P_h V_{h+1}^k - P_h V_{h+1}^\pi)(x_h^\tau, a_h^\tau) = (\phi_h^\tau)^\top \int (V_{h+1}^k - V_{h+1}^\pi) d\mu_h$, and the second uses $(\Lambda_h^k)^{-1} \sum_{\tau} \phi_h^\tau (\phi_h^\tau)^\top = I - \lambda(\Lambda_h^k)^{-1}$. Collecting the leading term of $\langle \phi, q_3 \rangle$ into Eq. (12) and defining

$$\Delta_h^k(x, a) := \langle \phi(x, a), q_1 \rangle + \langle \phi(x, a), q_2 \rangle - \lambda \left\langle \phi(x, a), (\Lambda_h^k)^{-1} \int (V_{h+1}^k - V_{h+1}^\pi) d\mu_h \right\rangle,$$

we obtain the claimed identity Eq. (10), since $\langle \phi(x, a), w_h^k \rangle - Q_h^\pi(x, a) = \langle \phi(x, a), w_h^k - w_h^\pi \rangle$. It remains to bound $|\Delta_h^k(x, a)|$. Each of the three contributions is of the form $\langle \phi(x, a), (\Lambda_h^k)^{-1} u \rangle$, which by Cauchy–Schwarz in the $(\Lambda_h^k)^{-1}$ inner product is at most $\|\phi(x, a)\|_{(\Lambda_h^k)^{-1}} \cdot \|u\|_{(\Lambda_h^k)^{-1}}$:

$$\begin{aligned} |\Delta_h^k(x, a)| &\leq \|\phi(x, a)\|_{(\Lambda_h^k)^{-1}} \left(\lambda \|w_h^\pi\|_{(\Lambda_h^k)^{-1}} + \left\| \sum_{\tau} \phi_h^\tau \varepsilon_h^\tau(V_{h+1}^k) \right\|_{(\Lambda_h^k)^{-1}} + \lambda \left\| \int (V_{h+1}^k - V_{h+1}^\pi) d\mu_h \right\|_{(\Lambda_h^k)^{-1}} \right) \\ &\leq \|\phi(x, a)\|_{(\Lambda_h^k)^{-1}} \left(\sqrt{\lambda} \|w_h^\pi\| + C dH \sqrt{\iota} + \sqrt{\lambda} \left\| \int (V_{h+1}^k - V_{h+1}^\pi) d\mu_h \right\| \right) \\ &\leq \|\phi(x, a)\|_{(\Lambda_h^k)^{-1}} \left(2H\sqrt{d} + C dH \sqrt{\iota} + 2H\sqrt{d} \right) \leq \beta \|\phi(x, a)\|_{(\Lambda_h^k)^{-1}}, \end{aligned}$$

where the first step is the triangle inequality and Cauchy–Schwarz on each term, the second uses $\|u\|_{(\Lambda_h^k)^{-1}} \leq \|u\| / \sqrt{\lambda_{\min}(\Lambda_h^k)} \leq \|u\| / \sqrt{\lambda}$ and the definition Eq. (7) of $\mathcal{E}$ (here the second term is exactly the quantity bounded by $C dH \sqrt{\iota}$ on $\mathcal{E}$), the third uses $\|w_h^\pi\| \leq 2H\sqrt{d}$ (Fact 1.4) and $\left\| \int (V_{h+1}^k - V_{h+1}^\pi) d\mu_h \right\| \leq H \|\mu_h(\mathcal{S})\| \leq H\sqrt{d} \leq 2H\sqrt{d}$, and the last absorbs the $\lambda = 1$ terms into the dominant $C dH \sqrt{\iota}$ and sets $C_\beta \geq C + 4$ so that $\beta = C_\beta dH \sqrt{\iota} \geq (C dH + 4H\sqrt{d})\sqrt{\iota}$. This completes the proof. $\square$

Lemma 3.2 (Optimism). *Under the hypotheses of Lemma 3.1, on the event $\mathcal{E}$ we have $Q_h^k(x, a) \geq Q_h^*(x, a)$ for all $(x, a, h, k) \in \mathcal{S} \times \mathcal{A} \times [H] \times [K]$, and hence $V_h^k(x) \geq V_h^*(x)$ for all (x, h, k) .*

Proof. Fix k and induct on h downward from $h = H + 1$ to $h = 1$, taking $\pi = \pi^*$ in Lemma 3.1. The inductive hypothesis at step $h + 1$ is $V_{h+1}^k(x) \geq V_{h+1}^*(x)$ for all x .

Base case $h = H + 1$. By convention $V_{H+1}^k \equiv 0 \equiv V_{H+1}^*$, so the hypothesis holds with equality.

Inductive step. Assume $V_{h+1}^k \geq V_{h+1}^*$ pointwise. By Lemma 3.1 with $\pi = \pi^*$ and the monotonicity of P_h ,

$$\begin{aligned} \langle \phi(x, a), w_h^k \rangle - Q_h^*(x, a) &= P_h(V_{h+1}^k - V_{h+1}^*)(x, a) + \Delta_h^k(x, a) \\ &\geq 0 - \beta \|\phi(x, a)\|_{(\Lambda_h^k)^{-1}} = -\beta \|\phi(x, a)\|_{(\Lambda_h^k)^{-1}}, \end{aligned}$$

where the first step is Eq. (10), and the second uses $P_h(V_{h+1}^k - V_{h+1}^*) \geq 0$ (from the inductive hypothesis and P_h preserving nonnegativity) together with $\Delta_h^k(x, a) \geq -\beta \|\phi(x, a)\|_{(\Lambda_h^k)^{-1}}$. Rearranging, $Q_h^*(x, a) \leq \langle \phi(x, a), w_h^k \rangle + \beta \|\phi(x, a)\|_{(\Lambda_h^k)^{-1}}$. Since also $Q_h^*(x, a) \leq H$, we conclude

$$Q_h^*(x, a) \leq \min \left\{ \langle \phi(x, a), w_h^k \rangle + \beta \|\phi(x, a)\|_{(\Lambda_h^k)^{-1}}, H \right\} = Q_h^k(x, a),$$

by the definition Eq. (4) of Q_h^k . Taking the maximum over a gives $V_h^k(x) = \max_a Q_h^k(x, a) \geq \max_a Q_h^*(x, a) = V_h^*(x)$, which advances the induction to step h . This completes the proof. $\square$

4 Potential lemmas

This section collects the three linear-algebraic and probabilistic facts the main proof invokes: a trace bound, the elliptical-potential (log-determinant) lemma, and the self-normalized concentration inequality. The first two are proved; the third is an external result cited from [1].

Lemma 4.1 (Trace bound). *Let $\Lambda_t = \lambda I + \sum_{i=1}^t \phi_i \phi_i^\top$ with $\phi_i \in \mathbb{R}^d$ and $\lambda > 0$. Then $\sum_{i=1}^t \phi_i^\top \Lambda_t^{-1} \phi_i \leq d$.*

Proof. We rewrite the sum as a trace and diagonalize. Let $\sum_{i=1}^t \phi_i \phi_i^\top = U \text{diag}(\sigma_1, \dots, \sigma_d) U^\top$ be an eigendecomposition, so that $\Lambda_t = U \text{diag}(\sigma_1 + \lambda, \dots, \sigma_d + \lambda) U^\top$. Then

$$\sum_{i=1}^t \phi_i^\top \Lambda_t^{-1} \phi_i = \sum_{i=1}^t \text{tr}(\Lambda_t^{-1} \phi_i \phi_i^\top) = \text{tr}\left(\Lambda_t^{-1} \sum_{i=1}^t \phi_i \phi_i^\top\right) = \sum_{j=1}^d \frac{\sigma_j}{\sigma_j + \lambda} \leq d,$$

where the first step uses $u^\top M u = \text{tr}(M u u^\top)$ and the cyclic property of the trace, the second uses linearity of the trace, the third substitutes the eigendecomposition (the matrix $\Lambda_t^{-1} \sum_i \phi_i \phi_i^\top$ has eigenvalues $\sigma_j/(\sigma_j + \lambda)$), and the last uses $\sigma_j/(\sigma_j + \lambda) \leq 1$ for each j since $\sigma_j, \lambda \geq 0$. This completes the proof. $\square$

Lemma 4.2 (Elliptical potential). *Let $\{\phi_j\}_{j \geq 1} \subset \mathbb{R}^d$ satisfy $\|\phi_j\| \leq 1$, let $\Lambda_0 \succeq I$ be positive definite, and set $\Lambda_j = \Lambda_0 + \sum_{i=1}^j \phi_i \phi_i^\top$. Then for every $t \geq 1$,*

$$\sum_{j=1}^t \phi_j^\top \Lambda_{j-1}^{-1} \phi_j \leq 2 \log \frac{\det \Lambda_t}{\det \Lambda_0} \leq 2d \log \frac{\lambda_{\max}(\Lambda_0) + t}{\lambda_{\min}(\Lambda_0)}. \quad (13)$$

In particular, if $\Lambda_0 = \lambda I$ with $\lambda \geq 1$, then $\sum_{j=1}^t \phi_j^\top \Lambda_{j-1}^{-1} \phi_j \leq 2d \log \frac{\lambda+t}{\lambda}$.

Proof. We first bound each summand by a logarithm, then telescope the determinants. Because $\lambda_{\min}(\Lambda_0) \geq 1$ and $\|\phi_j\| \leq 1$, the Mahalanobis length satisfies

$$\phi_j^\top \Lambda_{j-1}^{-1} \phi_j \leq \lambda_{\min}(\Lambda_{j-1})^{-1} \|\phi_j\|^2 \leq \lambda_{\min}(\Lambda_0)^{-1} \cdot 1 \leq 1,$$

where the first step is the Rayleigh bound, the second uses $\Lambda_{j-1} \succeq \Lambda_0$ and $\|\phi_j\| \leq 1$, and the last uses $\lambda_{\min}(\Lambda_0) \geq 1$. Using the matrix-determinant lemma $\det \Lambda_j = \det \Lambda_{j-1} \cdot (1 + \phi_j^\top \Lambda_{j-1}^{-1} \phi_j)$ and the elementary inequality $x \leq 2 \log(1+x)$ valid for $x \in [0, 1]$,

$$\sum_{j=1}^t \phi_j^\top \Lambda_{j-1}^{-1} \phi_j \leq \sum_{j=1}^t 2 \log(1 + \phi_j^\top \Lambda_{j-1}^{-1} \phi_j) = 2 \sum_{j=1}^t \log \frac{\det \Lambda_j}{\det \Lambda_{j-1}} = 2 \log \frac{\det \Lambda_t}{\det \Lambda_0},$$

where the first step applies $x \leq 2 \log(1+x)$ termwise, the second is the matrix-determinant lemma, and the third telescopes the logarithm. For the final bound, the eigenvalues of Λ_t sum to $\text{tr}(\Lambda_t) \leq \text{tr}(\Lambda_0) + t \leq d \lambda_{\max}(\Lambda_0) + t$, so by the AM–GM inequality $\det \Lambda_t \leq ((\lambda_{\max}(\Lambda_0) + t/d))^d \leq (\lambda_{\max}(\Lambda_0) + t)^d$, while $\det \Lambda_0 \geq \lambda_{\min}(\Lambda_0)^d$; substituting gives the right-hand inequality of Eq. (13). The case $\Lambda_0 = \lambda I$ follows by $\lambda_{\max}(\Lambda_0) = \lambda_{\min}(\Lambda_0) = \lambda$. This completes the proof. $\square$

Remark 4.3. Lemma 4.2 is the elliptical-potential (log-determinant) lemma of [1]; we reproduce the short proof for self-containment. The summand uses the *previous-time* matrix Λ_{j-1} , which is exactly the structure appearing in the LSVI bonus sum $\sum_k (\phi_h^k)^\top (\Lambda_h^k)^{-1} \phi_h^k$ of Section 5, since Λ_h^k is built from episodes $1, \dots, k-1$.

Finally, we restate the self-normalized concentration bound used in Lemma 2.2. It is an external result; we cite it rather than reproduce its proof.

Lemma 4.4 (Self-normalized bound, [1, Theorem 1]). *Let $\{\mathcal{F}_\tau\}$ be a filtration and $\{\eta_\tau\}$ a real-valued $\{\mathcal{F}_\tau\}$ -adapted sequence that is conditionally σ -sub-Gaussian: $\mathbb{E}[e^{s\eta_\tau} \mid \mathcal{F}_{\tau-1}] \leq e^{s^2\sigma^2/2}$ for all $s \in \mathbb{R}$. Let $\{\phi_\tau\} \subset \mathbb{R}^d$ be $\{\mathcal{F}_{\tau-1}\}$ -predictable, and set $\Lambda_t = \lambda I + \sum_{\tau=1}^t \phi_\tau \phi_\tau^\top$ with $\lambda > 0$. Then for any $\delta' \in (0, 1)$, with probability at least $1 - \delta'$, simultaneously for all $t \geq 0$,*

$$\left\| \sum_{\tau=1}^t \phi_\tau \eta_\tau \right\|_{\Lambda_t^{-1}}^2 \leq 2\sigma^2 \log \left(\frac{\det(\Lambda_t)^{1/2} \det(\lambda I)^{-1/2}}{\delta'} \right).$$

5 Regret bound

We now assemble the pieces into the regret bound. The proof conditions on the successful event $\mathcal{E}$, unrolls a per-step recursion, and splits the total into two named terms $T_1 + T_2$ (a bonus sum and a martingale term), closing the dominant term with the elliptical-potential lemma.

Theorem 5.1 (Sublinear regret of LSVI-UCB). *Suppose Assumption 1.1 holds. Run LSVI-UCB as in Eq. (2) through Eq. (5), with $\lambda = 1$ and $\beta = C_\beta dH\sqrt{\iota}$, where $\iota = \log(2dT/\delta)$, $T = KH$, and C_β is the constant of Lemmas 2.2 and 3.1. Then for any $\delta \in (0, 1)$, with probability at least $1 - \delta$,*

$$\text{Regret}(K) \leq C d^{3/2} \sqrt{H^3 T} \iota = \tilde{O}(d^{3/2} \sqrt{H^3 T}). \quad (14)$$

for a universal constant $C > 0$.

Proof. Throughout we condition on the event $\mathcal{E}$ of Lemma 2.2, on which all of Section 3 holds deterministically. Recall $\mathcal{E}$ has $\mathbb{P}[\mathcal{E}] \geq 1 - \delta/2$. Write $\phi_h^k := \phi(x_h^k, a_h^k)$ and define the per-step suboptimality and its martingale noise

$$\delta_h^k := V_h^k(x_h^k) - V_h^{\pi^k}(x_h^k), \quad \zeta_{h+1}^k := \mathbb{E}[\delta_{h+1}^k \mid x_h^k, a_h^k] - \delta_{h+1}^k. \quad (15)$$

Step 1: optimism reduces regret to $\sum_k \delta_1^k$. On $\mathcal{E}$, Lemma 3.2 gives $V_1^k(x_1^k) \geq V_1^{\pi^k}(x_1^k)$, so

$$\text{Regret}(K) = \sum_{k=1}^K [V_1^{\pi^k}(x_1^k) - V_1^{\pi^k}(x_1^k)] \leq \sum_{k=1}^K [V_1^k(x_1^k) - V_1^{\pi^k}(x_1^k)] = \sum_{k=1}^K \delta_1^k, \quad (16)$$

where the first step is the definition Eq. (1), the second uses optimism, and the third is Eq. (15).

Step 2: per-step recursion. On $\mathcal{E}$, applying Lemma 3.1 with $\pi = \pi^k$ at (x_h^k, a_h^k) and using $V_h^k(x_h^k) = Q_h^k(x_h^k, a_h^k)$ (greedy action) and $V_h^{\pi^k}(x_h^k) = Q_h^{\pi^k}(x_h^k, a_h^k)$,

$$\begin{aligned} \delta_h^k &= Q_h^k(x_h^k, a_h^k) - Q_h^{\pi^k}(x_h^k, a_h^k) \\ &\leq P_h(V_{h+1}^k - V_{h+1}^{\pi^k})(x_h^k, a_h^k) + 2\beta \left\| \phi_h^k \right\|_{(\Lambda_h^k)^{-1}} \\ &= \delta_{h+1}^k + \zeta_{h+1}^k + 2\beta \left\| \phi_h^k \right\|_{(\Lambda_h^k)^{-1}}, \end{aligned} \quad (17)$$

where the first step is Eq. (15) with the greedy/optimal action identities, the second uses $Q_h^k \leq \langle \phi, w_h^k \rangle + \beta \|\phi\|_{(\Lambda_h^k)^{-1}}$ from Eq. (4) and the recursion Eq. (10) (whose error term Δ_h^k is at most $\beta \|\phi_h^k\|_{(\Lambda_h^k)^{-1}}$, yielding the factor 2β), and the third uses $P_h(V_{h+1}^k - V_{h+1}^{\pi^k})(x_h^k, a_h^k) = \mathbb{E}[\delta_{h+1}^k \mid x_h^k, a_h^k] = \delta_{h+1}^k + \zeta_{h+1}^k$ by Eq. (15). Unrolling Eq. (17) from $h = 1$ to H with $\delta_{H+1}^k = 0$,

$$\delta_1^k \leq \sum_{h=1}^H \zeta_{h+1}^k + 2\beta \sum_{h=1}^H \left\| \phi_h^k \right\|_{(\Lambda_h^k)^{-1}}. \quad (18)$$

Step 3: two-term decomposition. Summing Eq. (18) over k and combining with Eq. (16),

$$\text{Regret}(K) \leq \underbrace{2\beta \sum_{k=1}^K \sum_{h=1}^H \left\| \phi_h^k \right\|_{(\Lambda_h^k)^{-1}}}_{T_1} + \underbrace{\sum_{k=1}^K \sum_{h=1}^H \zeta_{h+1}^k}_{T_2}, \quad (19)$$

where T_1 is the bonus sum, carrying the accumulated confidence width through the Gram matrix, and T_2 is the martingale term. We bound T_2 then T_1 .

Step 4: bound T_2 (martingale, Azuma–Hoeffding). By Remark 1.3, V_{h+1}^k is determined before the transition at episode k is revealed, so $\{\zeta_{h+1}^k\}$ is a martingale-difference sequence in the episode-step filtration with $|\zeta_{h+1}^k| \leq 2H$ (since $0 \leq \delta_{h+1}^k \leq H$). There are $KH = T$ terms, so by the Azuma–Hoeffding inequality, with probability at least $1 - \delta/2$,

$$T_2 = \sum_{k=1}^K \sum_{h=1}^H \zeta_{h+1}^k \leq 2H \sqrt{2T \log(2/\delta)} \leq 4H \sqrt{T} \iota, \quad (20)$$

where the first inequality is Azuma–Hoeffding with $N = T$ increments bounded by $2H$, and the second uses $\log(2/\delta) \leq \iota$ and $\sqrt{2} \leq 2$.

Step 5: bound T_1 (elliptical potential). By the Cauchy–Schwarz inequality applied to the inner sum over k , then the elliptical-potential lemma,

$$\begin{aligned} \sum_{k=1}^K \sum_{h=1}^H \left\| \phi_h^k \right\|_{(\Lambda_h^k)^{-1}} &= \sum_{h=1}^H \sum_{k=1}^K \left((\phi_h^k)^\top (\Lambda_h^k)^{-1} \phi_h^k \right)^{1/2} \\ &\leq \sum_{h=1}^H \left(K \sum_{k=1}^K (\phi_h^k)^\top (\Lambda_h^k)^{-1} \phi_h^k \right)^{1/2} \\ &\leq \sum_{h=1}^H (K \cdot 2d \iota)^{1/2} = H \sqrt{2dK} \iota, \end{aligned} \quad (21)$$

where the first step reorders the double sum, the second is Cauchy–Schwarz $\sum_k a_k \leq (K \sum_k a_k^2)^{1/2}$ with $a_k = \left\| \phi_h^k \right\|_{(\Lambda_h^k)^{-1}}$, and the third applies Lemma 4.2 with $\Lambda_0 = \lambda I = I$ and $t = K$ (note Λ_h^k is built from episodes $1, \dots, k-1$, matching the previous-time structure of the lemma), giving $\sum_{k=1}^K (\phi_h^k)^\top (\Lambda_h^k)^{-1} \phi_h^k \leq 2d \log(1+K) \leq 2d \iota$. Hence

$$T_1 = 2\beta \sum_{k,h} \left\| \phi_h^k \right\|_{(\Lambda_h^k)^{-1}} \leq 2\beta H \sqrt{2dK} \iota = 2C_\beta dH \sqrt{\iota} \cdot H \sqrt{2dK} \iota \leq 4C_\beta d^{3/2} H^2 \sqrt{K} \iota, \quad (22)$$

where the first equality is the definition of T_1 , the second substitutes Eq. (21), the third substitutes $\beta = C_\beta dH\sqrt{\iota}$, and the last collects constants using $\sqrt{2} \leq 2$.

Step 6: combine and convert to $T = KH$. Substituting Eq. (20) and Eq. (22) into Eq. (19), and using $H^2\sqrt{K} = \sqrt{H^4K} = \sqrt{H^3 \cdot KH} = \sqrt{H^3T}$ together with $H\sqrt{T} \leq \sqrt{H^3T}$ (as $H \geq 1$),

$$\text{Regret}(K) \leq 4C_\beta d^{3/2} H^2 \sqrt{K} \iota + 4H\sqrt{T} \iota \leq 4C_\beta d^{3/2} \sqrt{H^3T} \iota + 4d^{3/2} \sqrt{H^3T} \iota \leq C d^{3/2} \sqrt{H^3T} \iota,$$

where the first step substitutes the two term bounds, the second rewrites $H^2\sqrt{K} = \sqrt{H^3T}$ and bounds $H\sqrt{T} \leq d^{3/2} \sqrt{H^3T} \iota$ (using $d \geq 1$, $H \geq 1$, $\iota \geq 1$), and the last sets $C := 4C_\beta + 4$.

Step 7: union bound on the failure events. The bound above holds on $\mathcal{E} \cap \mathcal{E}'$, where $\mathcal{E}'$ is the Azuma event of Step 4. By Lemma 2.2, $\mathbb{P}[\mathcal{E}^c] \leq \delta/2$, and by Eq. (20), $\mathbb{P}[(\mathcal{E}')^c] \leq \delta/2$. By a union bound, $\mathbb{P}[\mathcal{E} \cap \mathcal{E}'] \geq 1 - \delta/2 - \delta/2 = 1 - \delta$. Therefore Eq. (14) holds with probability at least $1 - \delta$, which is $\tilde{O}(d^{3/2} \sqrt{H^3T})$ since $\iota = \log(2dT/\delta)$ is logarithmic. This completes the proof. $\square$

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